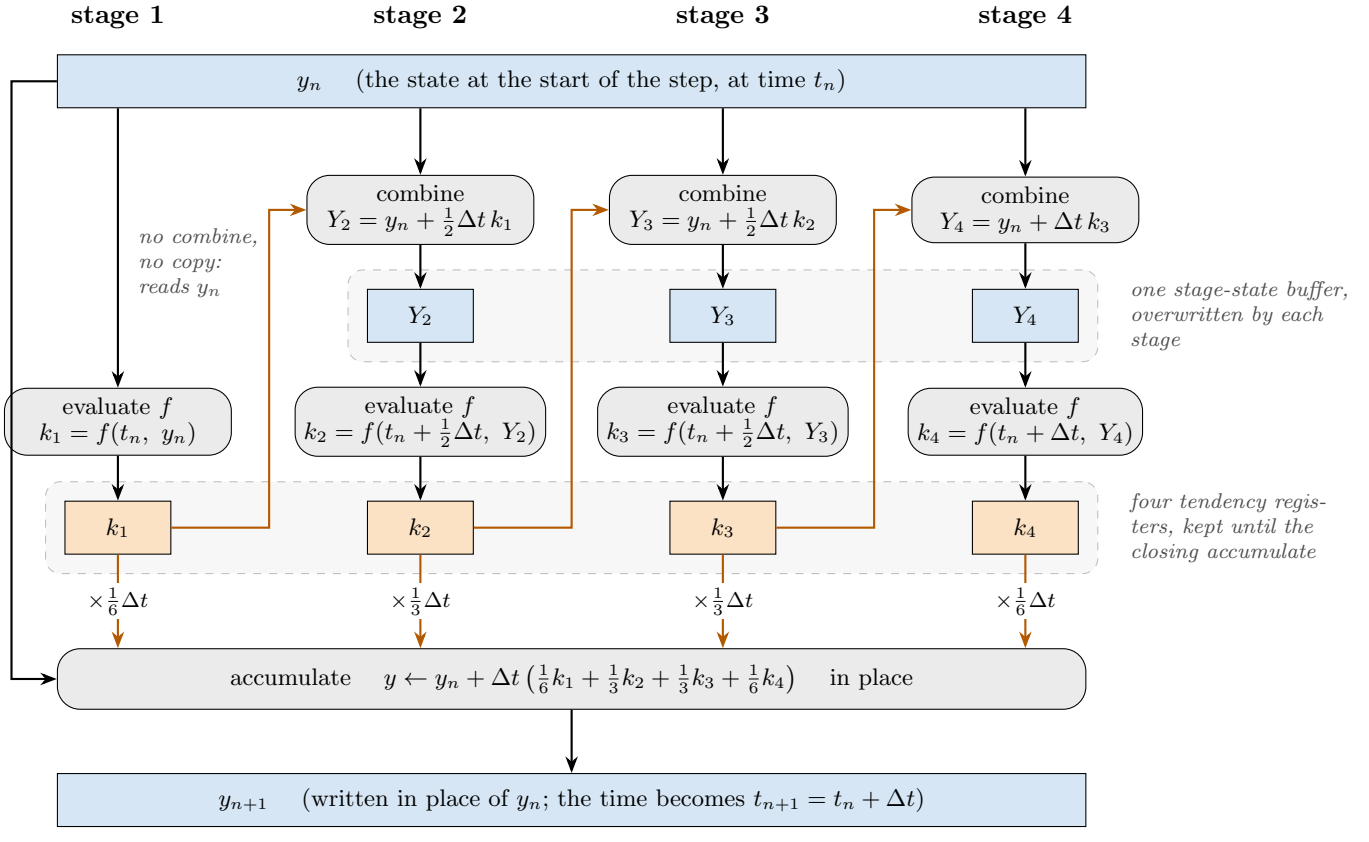

stage 1
stage 2
stage 3
stage 4
y_n (the state at the start of the step, at time t_n)
no combine, no copy: reads y_n
combine $Y_2 = y_n + \frac{1}{2}\Delta t\, k_1$
combine $Y_3 = y_n + \frac{1}{2}\Delta t\, k_2$
combine $Y_4 = y_n + \Delta t\, k_3$
Y_2
Y_3
Y_4
one stage-state buffer, overwritten by each stage
evaluate f $k_1 = f(t_n,\ y_n)$
evaluate f $k_2 = f(t_n + \frac{1}{2}\Delta t,\ Y_2)$
evaluate f $k_3 = f(t_n + \frac{1}{2}\Delta t,\ Y_3)$
evaluate f $k_4 = f(t_n + \Delta t,\ Y_4)$
k_1
k_2
k_3
k_4
four tendency registers, kept until the closing accumulate
$\times \frac{1}{6}\Delta t$
$\times \frac{1}{3}\Delta t$
$\times \frac{1}{3}\Delta t$
$\times \frac{1}{6}\Delta t$
accumulate $y \leftarrow y_n + \Delta t\left(\frac{1}{6}k_1 + \frac{1}{3}k_2 + \frac{1}{3}k_3 + \frac{1}{6}k_4\right)$ in place
y_{n+1} (written in place of y_n; the time becomes $t_{n+1} = t_n + \Delta t$)